

Application of Deep Learning for Solid Waste Trash Classification using Deep CNN

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Abstract- In the present era, with the rate of generation of wastes in concern, it is important to find an efficient solution to treat the wastes effectively. Segregation of wastes, being the most important part of waste management, is still done involving human labor. Our only hope to efficiently handle the wastes is to automate the segregation process and there arises a need for an efficient and robust classification model for segregating the wastes. This research proposes an advanced classification model DDR-net (Double fused Deep CNN using ResNext), which is an enhancement of the ResNext model boosted with double fusion and regularization. The proposed model can be implemented in real-time to any classification setup mechanism that feeds the trash image as input using a camera and based on the output from the model, actuators can be activated. The DDR-net classifies the solid wastes into their respective category with an accuracy of 97.81%. The performance of the proposed architecture was evaluated, experimented and compared with state-of-the-art architectures and it is observed that DDR-net outperforms the state-of-the-art methods on our dataset.

Keywords— Deep neural networks, Convolutional neural network, Deep learning, Fusion, Transfer learning, Waste classification.

I. INTRODUCTION

On the part of the local and urban bodies concerned the task of collecting, segregating and properly disposing of the solid wastes is a tedious task. Classification of solid wastes into biodegradable and non-biodegradable which is based on the nature of the solid waste is the key task when it comes to solid waste management. It takes lots of time and more manpower to classify the solid waste or in the worst case leaving the dump yards in dilemma. We are utilizing deep learning for accomplishing this task more efficiently. India, being a fast-growing economy with a massive population and expanding manufacturing infrastructure, discharges solid waste of 1.5 Metric Tons every day either into some water bodies by individuals and industries or in open lands. The existing techniques in solid waste management involve detecting the types of wastes using manpower and this solution is not efficient anymore as wastes are generated at an unmanageable rate and this concerns the wellbeing of our environment. Moreover, humans employed in the task of segregation are highly prone to many diseases.

Thus to automate this process with a classifier mechanism a highly efficient classification model is in need. Many methodologies were proposed in various literature involving various methodologies such as sensors, etc. to address this issue, which was not reliable and efficient to be implemented in real-time scenarios. Various machine learning-based approaches are also available in solid waste management but when implemented in real-time applications they fail to produce the desired level of accuracy. Attempting to solve this highly impacting problem and to benefit mankind with technology is the major motivation behind this research. The major traits of this research are summarized as follows:

1. Creation of dataset comprising 7000 images belonging to seven different classes namely, cardboard, paper, wood, glass, metal, plastic and e-waste. Among them, cardboard, paper and wood are Biodegradable wastes and glass, metal, plastic and e-waste are non-biodegradable wastes.
2. Experimentation with various deep neural architectures based on ResNet and playing around with the features extracted, with fusion mechanisms and harnessing the combined power of transfer learning along with double fusion.
3. DDR-net, an advanced and reliable classifier for solid waste classification, is developed, which can be implemented in any real-time mechanism for the classification of solid wastes.

The rest of this research proceeds as follows: Section 2 analyses many works of literature of the related work in this domain thoroughly. Section 3 describes the technology behind this research, while detailed explanations of the proposed methodology and its behavior are studied in section 4. Section 5 describes the preparation and preprocessing of data, followed by section 6 which suggests a methodology to prevent overfitting. Section 7 describes the learning methodology followed in this research and the details of the training environment are also provided here. The model is evaluated and analyzed in section 8 and the experimental setup including the work environment is also mentioned here. Finally, the experiments along with the

observed results and conclusions are presented in sections 9 and 10, respectively.

II. RELATED WORK

For image-based classification of solid wastes, many different algorithms namely Support Vector Machine (SVM), Artificial Neural Networks(ANN), Convolutional Neural Network(CNN), etc., have been developed in recent years. Chu, Y et al. [30] have studied a Multi-layer Hybrid Deep Learning System (MHS) to sort the waste thrown by individuals in public places. They have used a CNN-based algorithm and they were able to achieve classification accuracy levels of 90% and above. Bircanoğlu, C et al. [29] have developed a model namely RecycleNet, where they have carefully optimized deep CNN architecture for the classification of selected recyclable object classes. For training without any pre-trained weights, they were able to get the best accuracy of 90%, whereas, for transfer learning and fine-tuning of weight parameters, the best test accuracy achieved by them was 95%.

Hulyalkar, S et al. [12] have designed a system for automatic segregation of wastes into metal, plastic, glass and paper at the source itself. They have adopted a combination of CNN, image pre-processing and IoT. Sakr, G.E. et al. [15] have developed an automated system wherein waste materials like paper, metal and plastic can be segregated using CNN and Support Vector Machine (SVM) and made a comparison between these two techniques.

It is inferred from the above discussion that the Convolutional Neural Network surpasses most of the known algorithms in the task of image classification. Olugboja et al. [7] proposed an intelligent waste material classification system, which is developed using the 50-layer residual net pre-trained ResNet-50 model and Support Vector Machine (SVM). It was able to classify solid wastes like glass, metal, paper and plastic, etc. by achieving 87% accuracy on the trashnet dataset but the research has been carried out on a very small collection of trash image dataset containing only 1989 images of four different classes namely plastic, metal, glass, paper. In the research [14], Yu Liu et al. introduced and discussed the feasibility and superiority of Convolutional Neural Networks in the field of image fusion. They proposed a state-of-the-art CNN-based multi-focus image fusion method which also shows potential improvement in other-type image fusion methods.

In recent years many methods such as ResNet [2] and [8], ResNeXt [19], DenseNet121 [13], ImageNet [17] etc were introduced for classification using Convolutional Neural Networks (CNN). Saifuddin Hitawala et al. [19] implemented and evaluated the ResNext model architecture on the subset of the CIFAR-10 dataset that uses a homogeneous, multi-branch architecture for image classification. This research is evident to conclude that the ResNext model outplays most of the state-of-the-art models developed for image classification.

The proposed work is an effort made to make use of the benefits of multiple ResNext models by employing fusion mechanisms to develop a robust solid waste trash classification model using deep CNN (Convolution Neural Network)

III. ADVANCED SOLID WASTE CLASSIFICATION USING DEEP CNN

A. ResNext model

Residual Transformations Network for image classification, proposed by Xie et al. [27], is developed by utilizing the advantages of the split transform-merge strategy and the advantages of the residual block with extending its feature by the introduction of **cardinality**, which represents the number of different paths in the network. The ResNext model not only surpasses most of the state-of-the-art models but is also simple in terms of design when compared to other models like the Inception model, etc. Considering many such advantages and improvements, this research further proceeds with ResNext as base architecture.

B. Fusion methodology

Multiple deep CNN models have been developed since the emergence of CNN and deep learning and these models were able to achieve greater results and have been implemented to solve many problems. Fusion technology is based on the assumption that when multiple CNN models combined employing many fusion techniques, can yield promising results and can even outperform those individual deep CNN models. For improved and enhanced results, in early fusion, the features are extracted from images using multiple CNN models and these feature sets are fused, while the late fusion tries to fuse the classified outputs of multiple CNN models.

IV. PROPOSED METHODOLOGY

Here, the research involves utilizing the combined advantages of a fine-tuned ResNext-101 model and a Resnext-50 model fully trained from scratch, on our dataset, by a double fusion process. Here, the features are extracted from the input images, using both the models separately and the early fusion process takes place. Early fusion of the obtained feature vectors from both the models is fused by DCA followed by simple concatenation [2]. Then features undergo classification using the softmax layer and the outputs from both the models are fused using PSO-based weight optimization, a late fusion method. Then, at last, the result of early fusion and late fusion undergo Double fusion. The following section describes further the above-mentioned processes thoroughly.

A. Phase 1- Extracting features and Classification:

The research mainly aims to harness the power of double fusion. Inferred from the research [27], ResNext is chosen as the base model to proceed with and we use two models for

feature extraction, where both of them are different versions of ResNext. Both the models are used to extract features from the images upon which both early fusion and late fusion are to be employed at later phases. The overview of the proposed methodology is given by the fig 1. The extracted features are fed into their respective classifier layers (log softmax layers) and the features are classified accordingly. The two models used are as follows:

1. The first CNN model is a pre-trained ResNext-101 model with cardinality $C=32$, enhanced by fine-tuning. As [3] suggests, the bottom layers are better for feature extraction as they are generic in nature and thus we use them to extract the features from the images, upon which early fusion is to be employed. Then the model's fully connected layers are removed and four new fully connected layers are added to the network, which is used for the classification of the features.

2. The second CNN model in the proposed methodology is a ResNext-50 model, with cardinality $C=32$, which is fully trained from scratch on our dataset and the features are extracted using the model, which is fed to the classifier.

B. Phase 2- Early fusion :

The features extracted from both the models discussed above in phase 1 are fused using the Discriminant Correlation Analysis (DCA) method. As the research [4] suggests, the DCA is employed for the early fusion of the features extracted from the models, as it vanquishes the problems faced in Canonical Correlation Analysis (CCA). Here the feature vector of length 1000, from both the models are extracted from the fully connected layer (FC-1000). Once, the feature sets from both the models are processed with DCA, they are combined using the summation process [27]. After fusion takes place, the fused feature set is fed into the classifier and the classifier predicts the output based on the fused feature set.

C. Phase 3- Late Fusion:

The classified outputs corresponding to each model, obtained from phase 1 are fused using the late fusion method: Particle Swarm based optimization of the weights (PSO method), as PSO is easy and flexible to implement as it combines both social concepts [6] and swarm intelligence principles. Here, the accumulation accuracy (1) on the validation set, that represents the weights to be assigned, is computed as Eq. 1.

$$A_{acc} = x(1) * P_1 + x(2) * P_2 + \dots + x(n) * P_n \quad (1)$$

where, P_n represents the probabilities obtained through the n th model and $x(n)$ represent the value (weight) to be used for the model.

D. Phase 4- Double fusion:

Once both the predictions from phase 2 and 3 are calculated, they undergo the double fusion based on the accumulative accuracy represented by Eq. 2.

$$A_{acc} = x(1) * P_{early\ fusion} + x(2) * P_{late\ fusion} \quad (2)$$

where, $P_{early\ fusion}$ and $P_{late\ fusion}$ represent the posterior probabilities obtained with early and late fusion on the validation set, respectively.

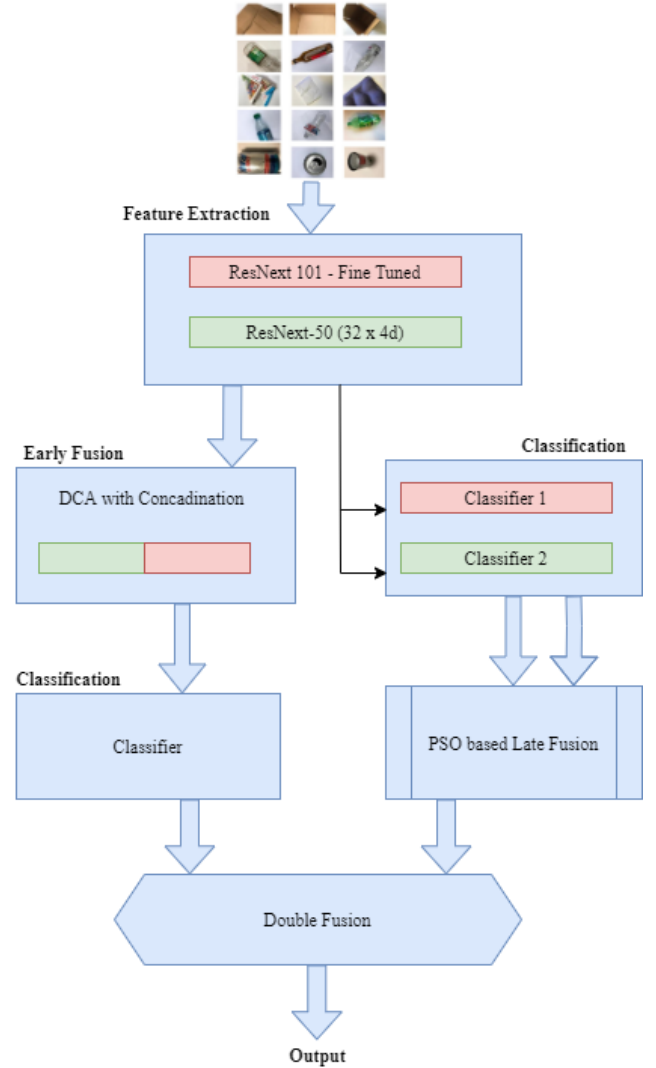


Fig.1 Overview of the proposed methodology

V. EXPERIMENTAL DATASET

The dataset used in this work consists of:

1. Images from Kaggle (www.kaggle.com/dimonisochecolo/trash-dataset).
2. Images from GitHub (<https://github.com/garythung/trashnet>).
3. Images collected from Google and Bing searches.
4. Images captured manually under a controlled environment.

The dataset consists of 7000 images split equally into seven categories namely cardboard, paper, wood, e-waste, glass, metal and plastic. The images of type[I] and [II], being synthesized, don't require much preprocessing, but the

images of type [III] and [IV] are preprocessed by resizing and cropping into optimal sizes if needed. The unwanted noise in the images has been filtered and removed for better accuracy.

Table 1. Bio Degradable

Cardboard	Paper	Wood
		

Table 2. Non Bio Degradable

E - waste	Glass	Metal	Plastic
			

VI. AVOIDING OVERFITTING

As the trainable parameters of deep CNN are very huge in number and there needs the care to be taken during the training phase and the most important concern is that there is a high risk of overfitting, if the data available is relatively inadequate. The following measures are taken in this work to prevent overfitting from happening. An attempt has been made to enhance the dataset during the training phase, using data augmentation techniques like the creation of transformations of zooms, flips, shifts of images, etc. as represented in figure 2.

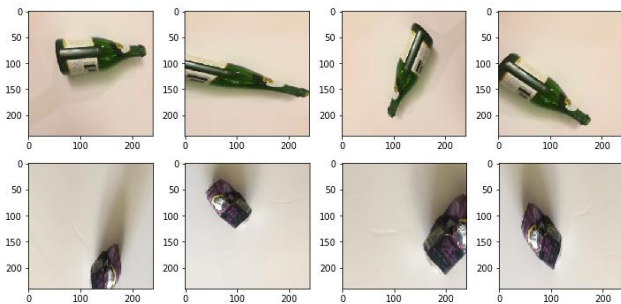


Fig.2 Results from employing various augmentation techniques on training images.

The most efficient step in preventing overfitting is regularization, which penalizes the model through the addition of some penalty to the loss function and this helps in compensating the unavailability of huge data. So, inferred from [31], L2 regularization along with dropouts of rate $p=0.4$ [11] has been employed. The research incorporates the above-discussed regularization methods to prevent overfitting.

VII. LEARNING METHODOLOGY AND TRAINING ENVIRONMENT

The proposed methodology was implemented using python 3.7 with Pytorch, an open-source deep learning framework. The model was trained on NVIDIA Tesla P100 PCIe 16 GB GPU using Ubuntu 16.04LTS.

Both the ResNext models were trained using the stochastic gradient descent (SDG) optimizer, with a batch size of 256 images and with a learning rate, $\alpha=0.0001$. A weight decay of 0.001 has been used to reduce the training error. Adam optimizer with $\alpha=0.001$ has been used for the base network in model 1 and SDG is used for fine-tuned layers of model 1, while model 2 is fully trained using SDG optimizer. Both the models are trained for 50 epochs. The maximum number of stall iterations was kept at 10 and the swarm size was fixed as 20 for the fusion techniques. The results are tabulated as a confusion matrix in table 3.

Further, the performance of the proposed model must be compared with the state-of-the-art models to understand and interpret how better the proposed model performs. For this purpose, this research utilized the architecture DNN-TC proposed by [2], which is a version of ResNext-101, a state-of-the-art model, enhanced with predictive performance. DNN-TC was run for 50 epochs (suggested by [2]) on our dataset, using SDG optimizer with $\alpha=0.0001$ and the results are tabulated in table 4.

VIII. EVALUATION METRICS

Table 3. Confusion matrix of the proposed method (DDR-net)

	Actual positive	Actual negative
Predicted positive	602	10
Predicted negative	13	425

Table 4. Confusion matrix of state-of-the-art model (DNN-TC)

	Actual positive	Actual negative
Predicted positive	514	34
Predicted negative	36	446

Table 5. Comparison of evaluation metrics

	DDR-net	DNN-TC
Accuracy	0.9781	0.9333
F1 Score	0.9813	0.9362
Matthews Correlation Coefficient	0.9549	0.8664
Precision	0.9837	0.9380
Recall	0.9788	0.9345

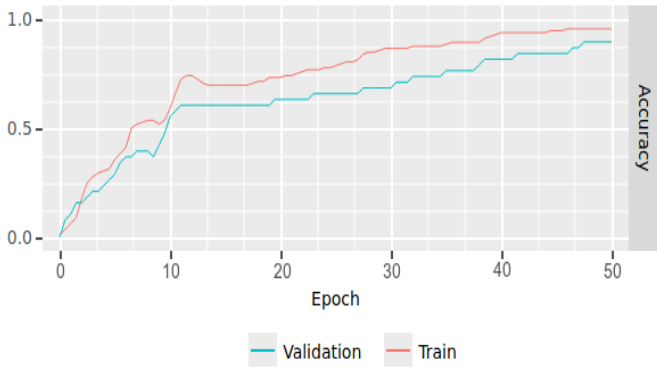


Fig 3. Training and validation accuracy of the proposed model

IX. RESULTS

Here, the detailed interpretation of the evaluations of the proposed work and also the state-of-the-art method is discussed. Table 5 displays the confusion matrix of our proposed work from which it is evident that our proposed model (DDR-net) classifies the solid wastes with an accuracy of 97.3%, which is a great improvement over the DNN-TC, implemented on our dataset with an accuracy of 93%. Further, table 5 represents the comparison of other evaluation metrics between the proposed model and the existing state-of-the-art model. Figure 3 represents the training-validation accuracy graph of the proposed methodology respectively. By inspecting table 5, it is concluded that the proposed model of this research performed better on our dataset and the double fusion method along with data augmentation and regularization has contributed well to the efficiency of the model and the results are promising, as the model outperformed the DNN-TC, while implemented on our dataset.

X. CONCLUSION

An advanced deep CNN, DDR-net with the combined advantage of residual connections and double

fusion, which is robust, efficient and reliable in the task of classification of solid wastes has been proposed in this research. After analyzing various works of literature and understanding the advancement in deep CNN models, an attempt has been made here, through our experiments, to justify and support the idea that more than one deep CNN models synthesized together by double fusion of features of individual models, can have improved efficiency and can even outperform the state-of-the-art models in the task of classification of solid wastes.

This work can further be enhanced by attempting to increase the size of the dataset by collecting more valid and relevant data, by tuning various hyperparameters. In future work, a real-time implementation of the model in an IoT-powered bin kind of structure is planned to be developed, that is suitable in both indoor and outdoor environments, capable of segregating the solid wastes based on the model's output.

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